Rows Columns and Data Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected schema and validation record.

Specify duration fields, test boundaries, convert units and separate a valid row from a true fact.

Keep one habit: Keep the contract, classification reasons, original source and repair note together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Debugging Strategies

Trace Tables

Precision in Algorithms

W1 · Write the duration contract

> Include row meaning, fields, types, units and inclusive boundaries.

Your evidence record

- Use the classroom-duration contract for W1–W3 and W5: one session per row; activity is nonempty text; minutes is an exact integer from 0 through 60 inclusive, measured in minutes. For W2 assume activity="Reading" and both fields are present. A source-verified unit conversion is a documented repair, not permission to guess a duration.

- List the required fields, **type**s and duration boundaries for the worked table.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Classify four boundary records

> Give the deciding rule for each value.

Your evidence record

- Duration reference: activity is nonempty text and minutes is an exact integer 0–60 inclusive. Both fields are required; one row represents one activity session. For W2, activity="Reading" and both fields exist.

- Classify minutes values 0, 60, -1 and 61 under this **schema**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Document a verified unit conversion

> Keep the original source value and conversion note.

Your evidence record

- Duration reference: activity is nonempty text and minutes is an exact integer 0–60 inclusive. Both fields are required; one row represents one activity session. For W2, activity="Reading" and both fields exist.

- A source says 180 seconds but the row says minutes: 180. Give a documented correction.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Limit a validation claim

> Explain which evidence is still missing.

Your evidence record

- Duration reference: activity is nonempty text and minutes is an exact integer 0–60 inclusive. Both fields are required; one row represents one activity session. For W2, activity="Reading" and both fields exist.

- A row says Reading, 12 and passes. Does that establish the session really lasted **12 minutes**?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Design a schema independently

> Keep one first attempt before hints or feedback.

Read and explain independently

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Design a schema independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Store one book per row and keep reader preferences in a separate record.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Classify records for Rows **column**s and **schema**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Rows **column**s and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-**input** case.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended **type**s and a repair.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A table row is book_id B7, pages 84, available true. Name the **type** of each field and explain why pages stored as eighty-four complicates calculation.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 7 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-**input** case.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

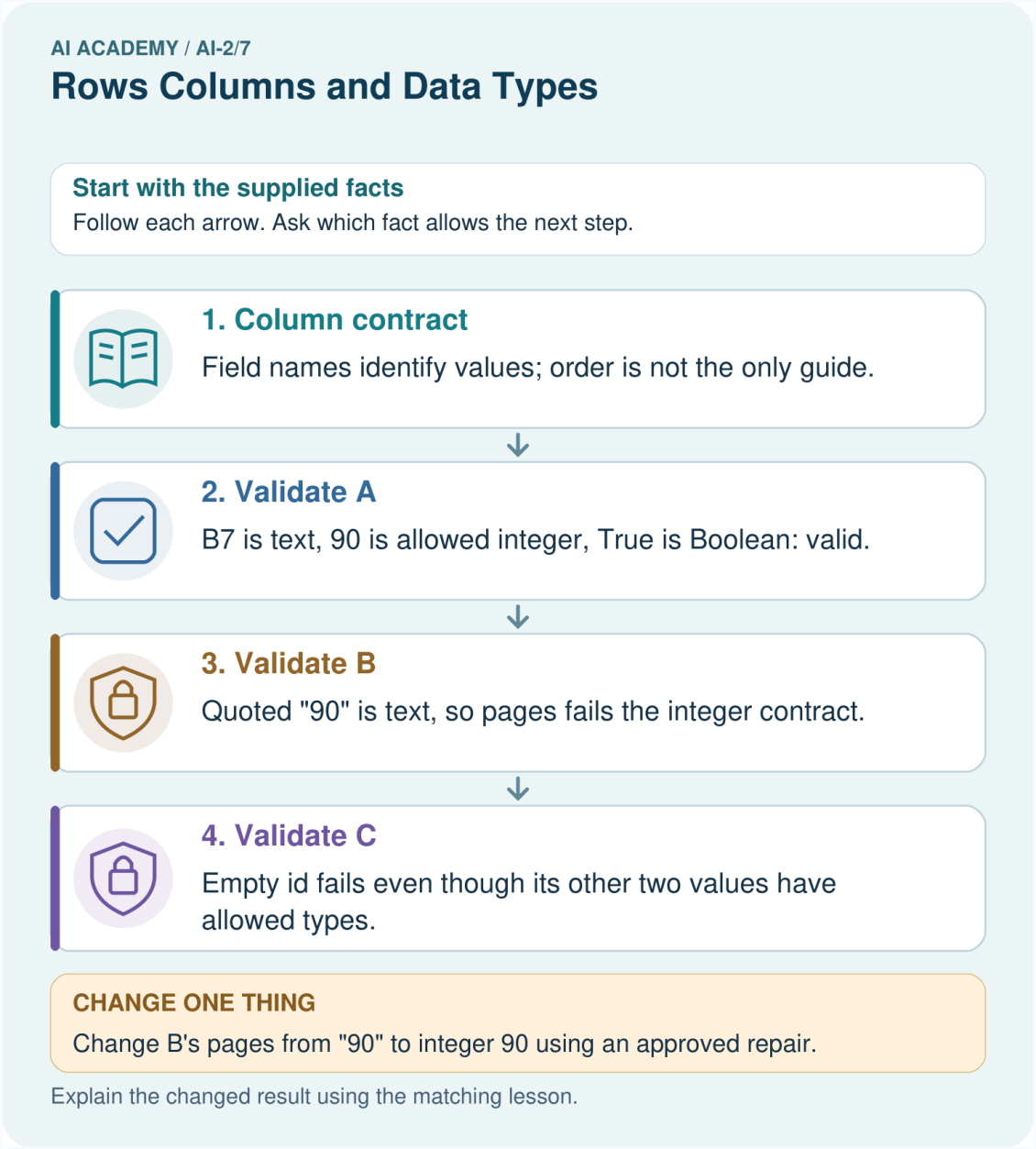
> Keep the explanation and the support you used.

Use this when helpful

A fictional approved book catalogue has three fields per row.

Schema: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which two rows originally fail, and do they fail for the same reason?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional approved book catalogue has three fields per row.
- Which labelled step matches this detail? Empty id fails even though its other two values have allowed **type**s.
- 1. **Column** contract
- 2. Validate C

- 3. Validate B
- 4. Validate A
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Validate C Empty id fails even though its other two values have allowed **type**s. A **schema** makes row meanings and expected **type**s explicit.

The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source.

Changed-case question

- Changed-case practice: Change B's pages from "90" to integer 90 using an approved repair. Which conclusion is supported by the supplied facts?
- 1. B now satisfies the declared **schema**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: B now satisfies the declared **schema**. Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Repair and save your work

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

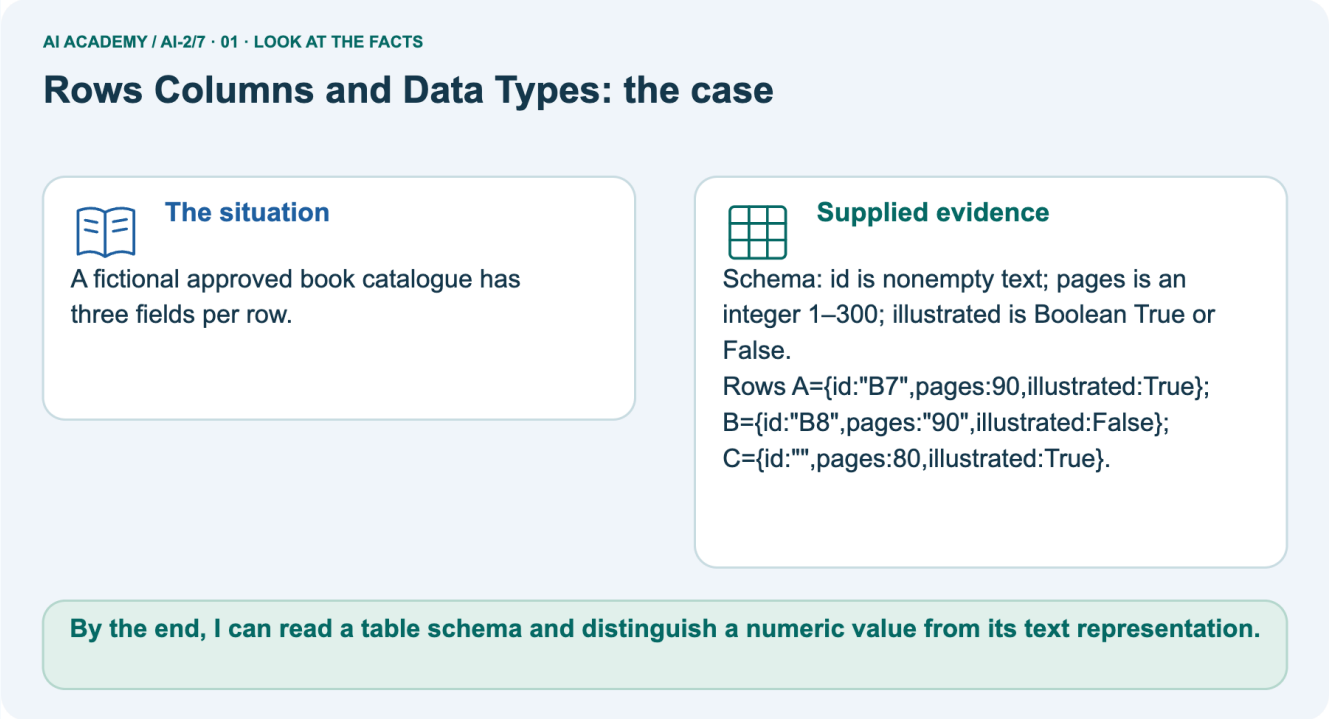
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Rows Columns and Data Types: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

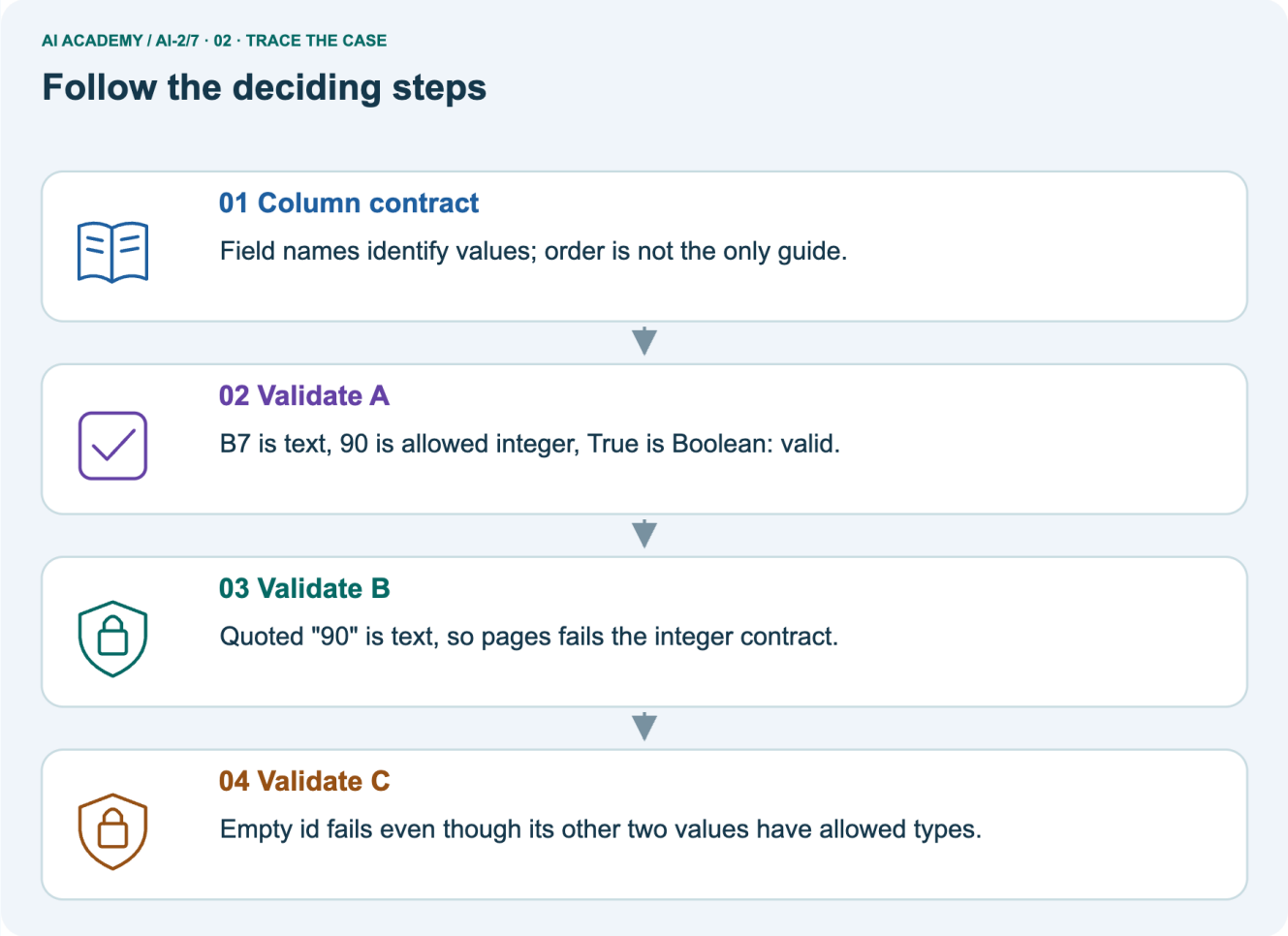
A fictional approved book catalogue has three fields per row. Supplied facts: **Schema**: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



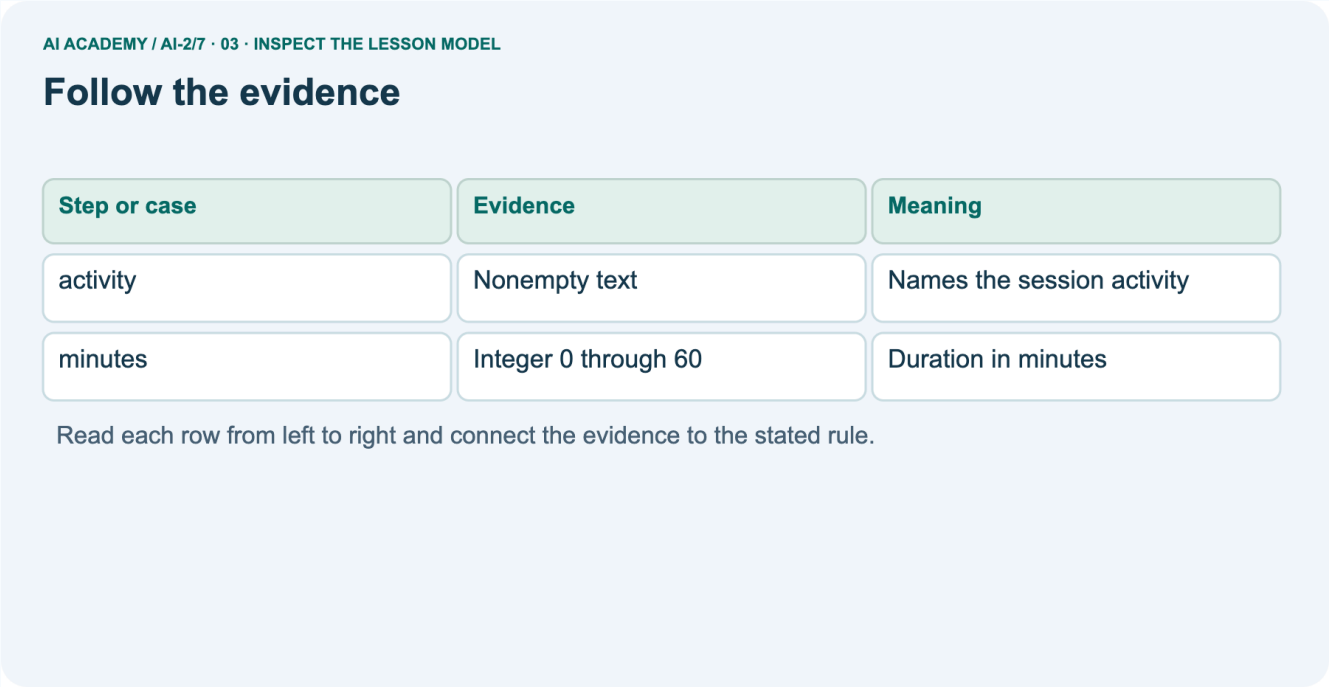
Follow the labelled picture with the matching explanation. Open the vector diagram

Column contract: Field names identify values; order is not the only guide. → Validate A: B7 is text, 90 is allowed integer, True is Boolean: valid. → Validate B: Quoted "90" is text, so pages fails the integer contract. → Validate C: Empty id fails even though its other two values have allowed **type**s.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



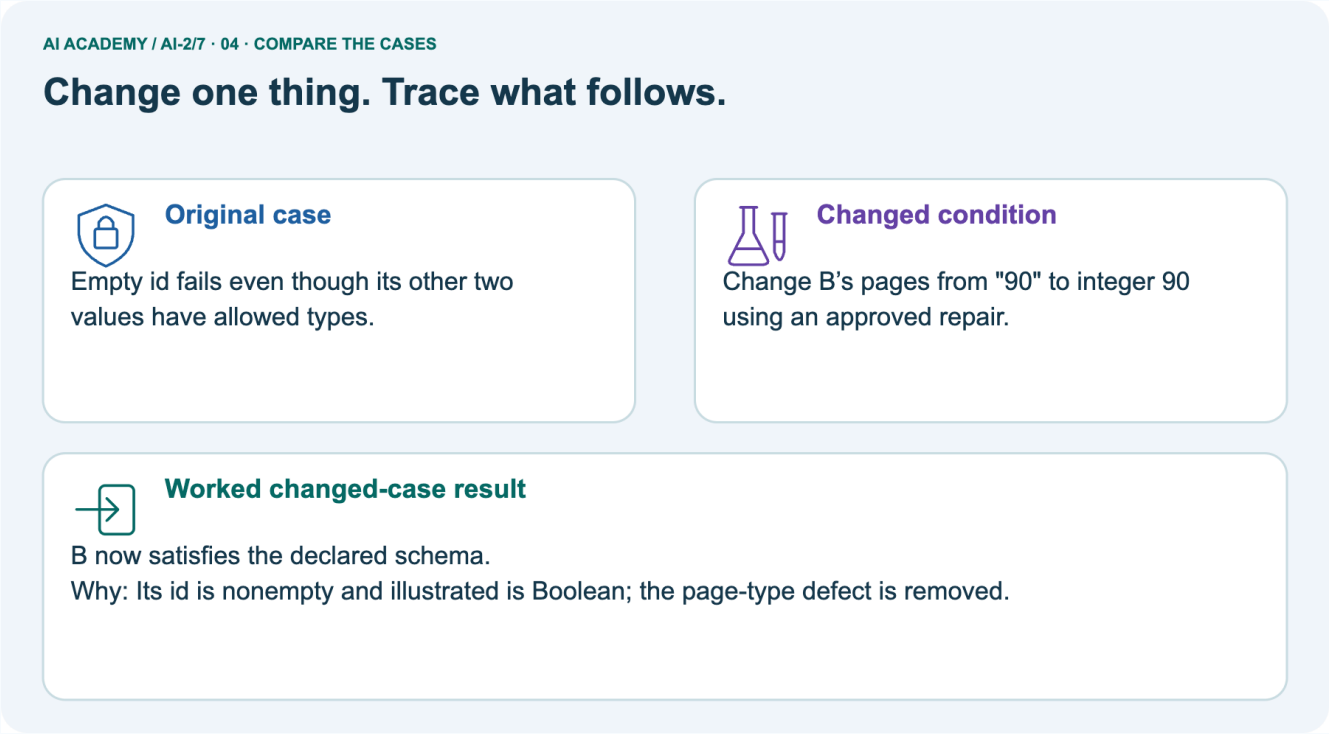
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

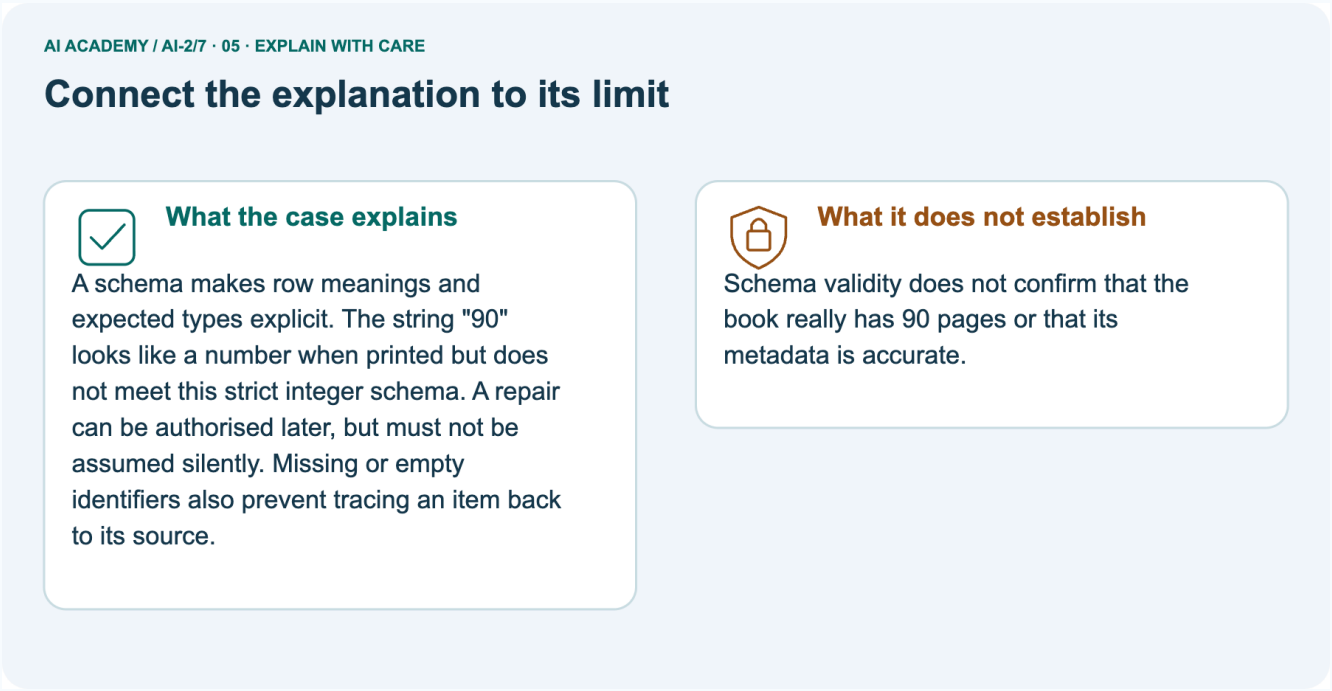
Original case: Empty id fails even though its other two values have allowed **type**s. Changed condition: Change B's pages from "90" to integer 90 using an approved repair. Expected result: B now satisfies the declared **schema**.

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

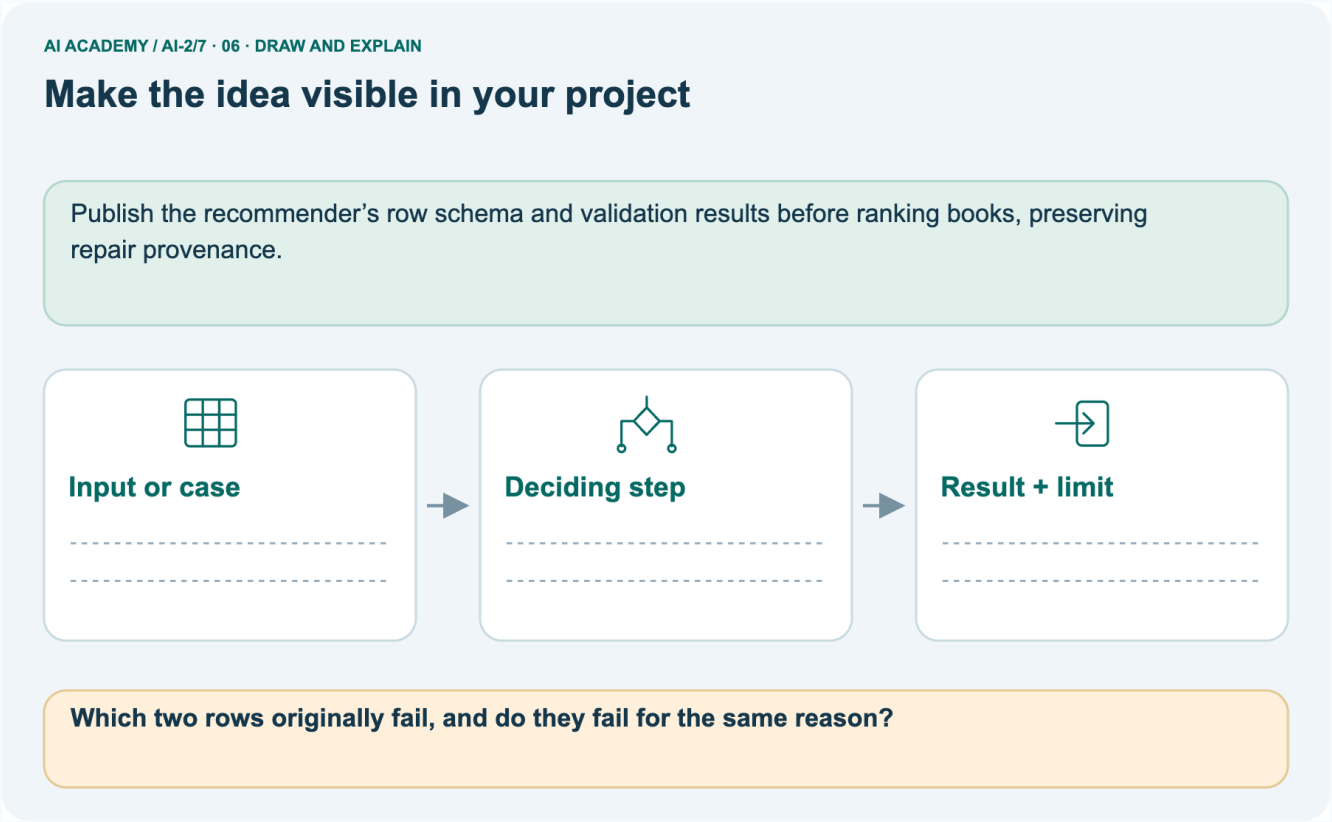
A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source. Limit: **Schema** validity does not confirm that the book really has 90 pages or that its metadata is accurate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which two rows originally fail, and do they fail for the same reason?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Rows Columns and Data Types: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.